

Systems Under Pressure

Shared Group Investigation Notebook

DRIVING QUESTION

How do we know when a system is actually in danger?

GROUP DETAILS

GROUP NAME _____

MEMBERS _____

INVESTIGATION TOPIC _____

CATEGORY _____

ESSENTIAL QUESTION _____

HOW THIS NOTEBOOK WORKS

- One shared investigation journal per group.
- Individual handwritten extensions are separate and shown during vlogs.
- Three informal vlogs: sequences, exponentials, polynomials.
- Short handwritten reflections live inside the individual extensions.

CORE ROUTINES

01 SYSTEM -> VARIABLES -> MODEL -> CRITIQUE

02 THINK FIRST -> SPARKY -> VERIFY -> DECIDE

03 LOCAL CHECK vs GLOBAL CHECK

04 WHAT THE MODEL REVEALS / HIDES

05 ONE REJECTED MODEL

THE WORK OF THIS NOTEBOOK

Graphs, equations, calculations, annotations, revision, evidence, and one rejected model.

Before building any model, understand the situation: who is affected, what they experience, what the data shows, and what it hides.

MAP

Use this page for free exploration: what do we know, what surprises us, what makes this urgent, and what investigation question should we draft?

- > Initial sense of the topic and draft investigation question.
- > At least 2 stakeholder perspectives mapped.
- > Connected systems map with one confounding variable named.
- > One possible source or graph identified.

OPEN WORKSPACE: notice, question, sketch, list, connect

Use this page to identify who is affected, what they experience, what variables they would track, and where perspectives conflict.

SYSTEM	VARIABLES	MODEL	CRITIQUE
Stakeholders Who is affected, and how?	Variables What would each person measure or care about?	Conflict Where do perspectives disagree?	
OPEN WORKSPACE			

Use this page to map connected variables, possible hidden pressures, measurable evidence, and at least one confounding variable.

OPEN WORKSPACE

Graph Observations

Use this page to annotate trends, thresholds, anomalies, and intervals before choosing equations.

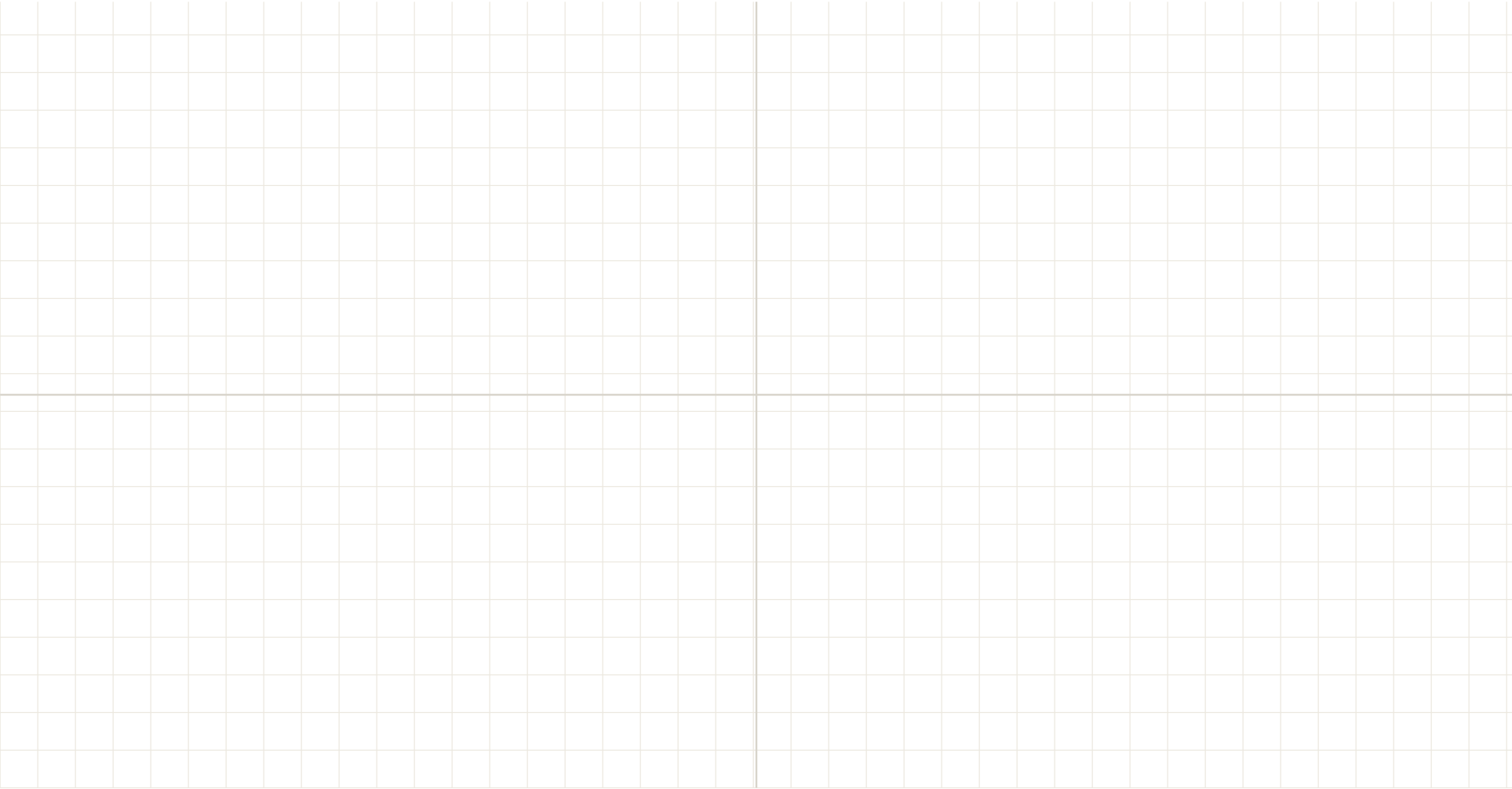
SYSTEM	VARIABLES	MODEL	CRITIQUE
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Graph First
What feature matters most?

Interval
Where does the pattern change?

Evidence
Which part supports your claim?

OPEN WORKSPACE



Local vs Global Behavior

Use this page to compare a short interval with a longer interval and notice model failure.

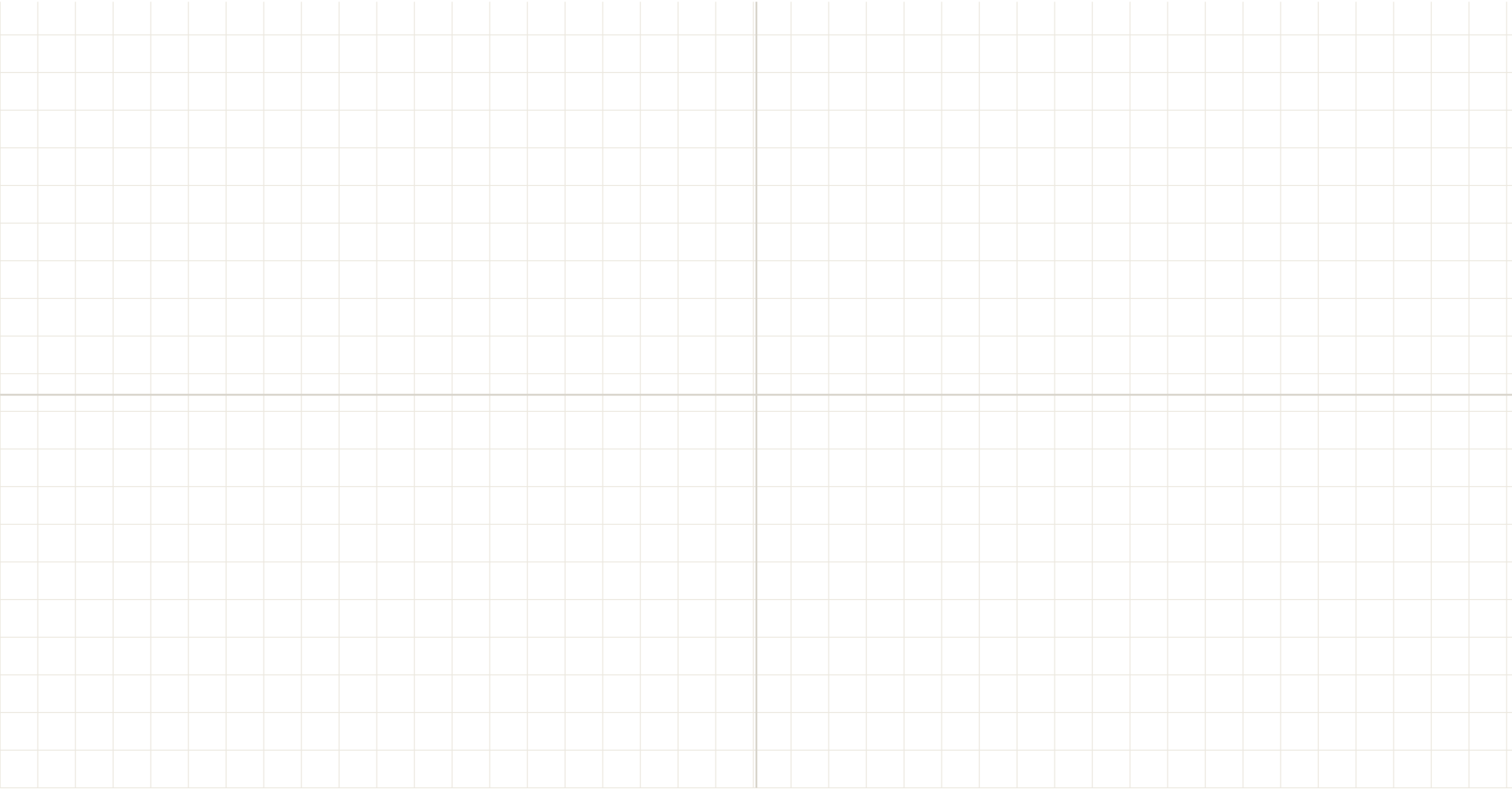
SYSTEM	VARIABLES	MODEL	CRITIQUE
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Zoom In
What seems true here?

Zoom Out
Does it still make sense globally?

Bad Extrapolation
Where did the model stop being meaningful?

OPEN WORKSPACE



Sequences + Series Modeling

Use this page to choose a changing pattern, test same amount vs same ratio, model accumulation, and critique where the pattern breaks.

SYSTEM	VARIABLES	MODEL	CRITIQUE
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System / Variables

What is changing, and which two values anchor the pattern?

recursive $a_n = r \cdot a_{n-1} + d$

Model

Sequence, recursion, explicit rule, or partial sum.

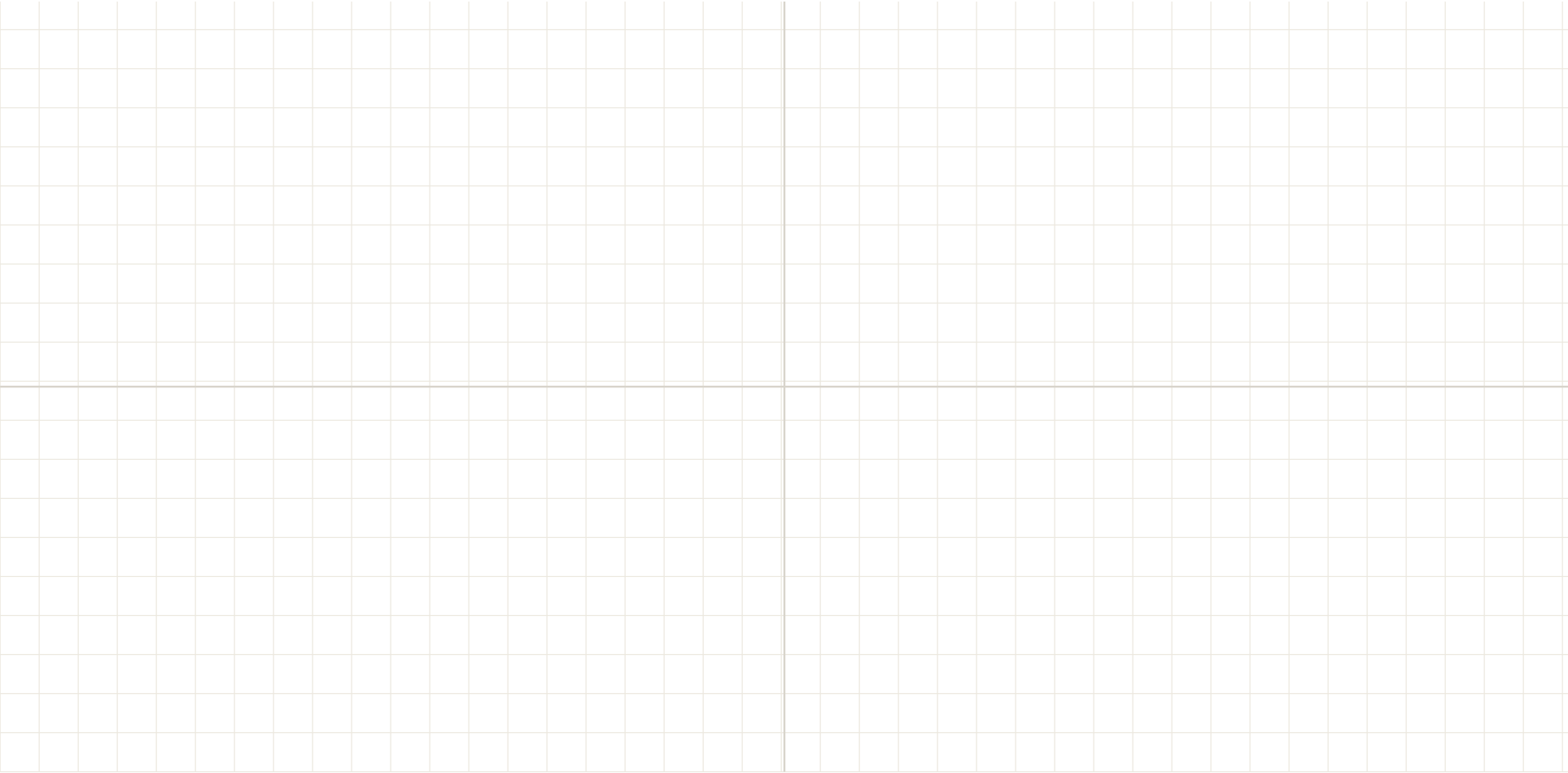
explicit $a_n = a_1 r^{n-1}$

Critique

What does the math reveal, and where does it stop working?

series $S_n = a_1 + a_2 + \dots + a_n$

OPEN WORKSPACE



Exponentials + Thresholds

Use this page to test repeated percent change, solve for an important threshold, and decide where exponential reasoning fails.

SYSTEM	VARIABLES	MODEL	CRITIQUE
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System / Variables

What quantity changes by percent or ratio?

model

$$f(x) = a \cdot b^x$$

Threshold

When does the model reach an important system value?

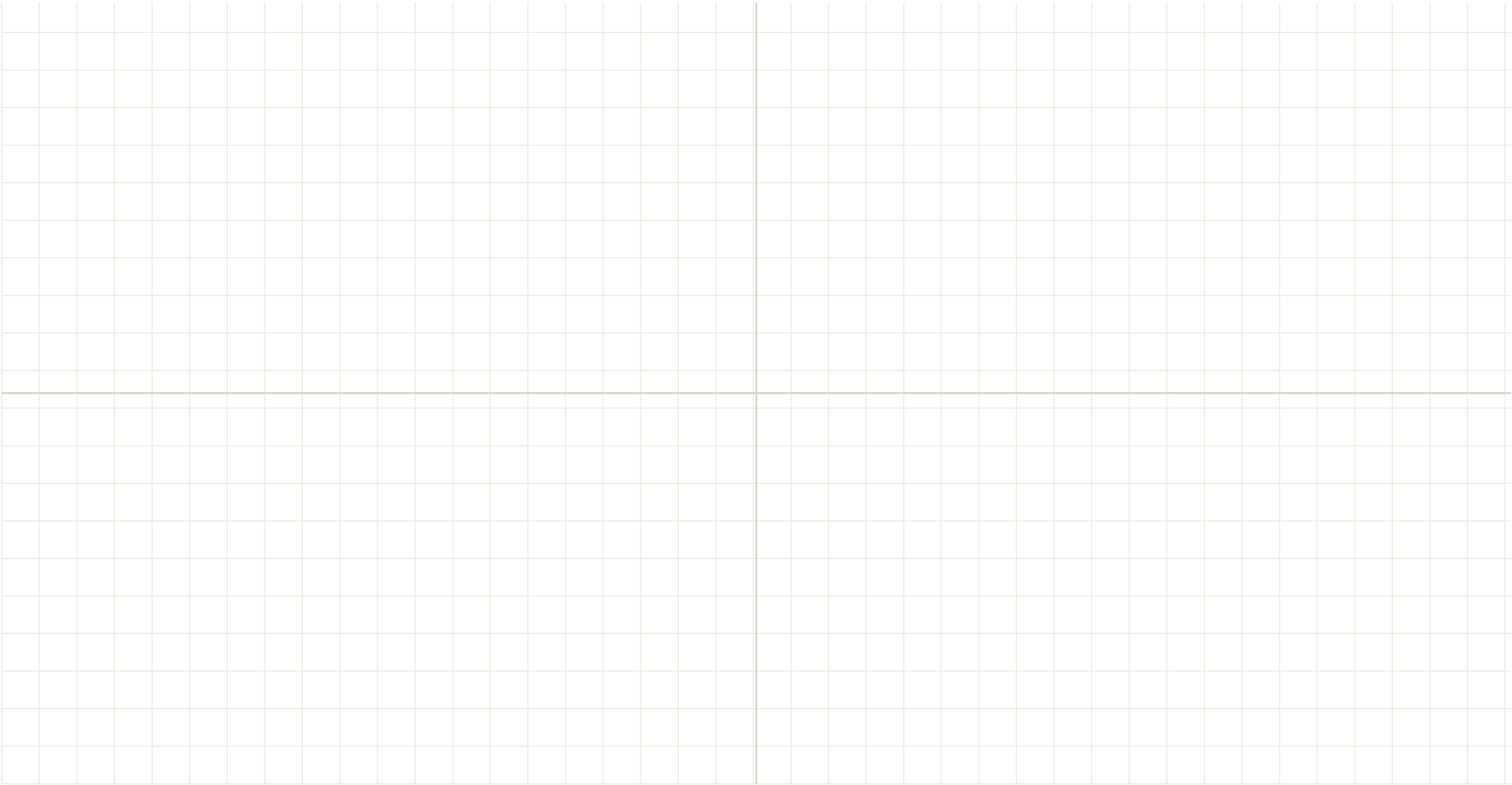
solve

$$\log_b(\text{threshold}/a) = x$$

Critique

What happens if you extend the model too far?

OPEN WORKSPACE



Polynomial Behavior

Use this page to study intervals where behavior changes direction, crosses or bounces off thresholds, or looks different locally than globally.

SYSTEM	VARIABLES	MODEL	CRITIQUE
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System / Variables

Which interval shows a change in behavior?

factored

$$P(t) = a(t-r_1)(t-r_2)^2...$$

Model

Turning points, thresholds, increase/decrease, cross/bounce.

zero

$$P(t)=0$$

Critique

Where does polynomial end behavior stop making sense?

feature

multiplicity: cross or bounce?

OPEN WORKSPACE



Use this page to compare models by what they predict, reveal, hide, assume, and fail to explain.

OPEN WORKSPACE

Confounding Variables

Use this page to separate visible correlation from possible causes and alternative explanations.

SYSTEM	VARIABLES	MODEL	CRITIQUE
Related What evidence suggests connection?	Not Proof What else could explain it?	Sentence Frame These variables appear related because ____, but this does not prove causation because ____.	
OPEN WORKSPACE			

Use this page to keep notebook, extensions, and vlogs connected. Sparky coaches the process; it does not complete the assignment.

OPEN WORKSPACE

Use this page to show how evidence, critique, or model failure changed your group's interpretation.

OPEN WORKSPACE

Use this page to make a defensible claim with graph evidence, calculation evidence, rejected-model evidence, and uncertainty.

OPEN WORKSPACE

Use this page to prepare a mathematical argument: strongest evidence, biggest limitation, rejected model, and final defense.

Explain	Defend	Close
Every graph and equation verbally.	Mathematical, realism, assumption, interval, counterargument tests.	Not a summary. A mathematical argument.

Use this page for last annotations, slide notes, and questions your group must be ready to answer.

Ready All members can explain the math.	Artifacts Notebook, vlogs, extensions, and sources are ready.	Attend Presentation day is during class on Tuesday, June 9.
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